

Marcos Treviso

Curriculum Vitae

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📄 <https://mtreviso.github.io>

Google Scholar

Professional

Position. Assistant Professor at the Computer Engineering Department at Instituto Superior Técnico, University of Lisbon. I am also a member of ELLIS.

Research. I work on efficiency, long-context modeling, and interpretability in ML & NLP.

Post-doc. I was a post-doc researcher at Instituto de Telecomunicações during 2024-2025.

Education

2019–2023 **PhD in Electrical and Computer Engineering**, *Instituto Superior Técnico*, University of Lisbon, Portugal.

Selective Sparsity for Explainability with Applications to Translation Quality Estimation

**With Distinction and Honour.*

Supervisor Dr. André F. T. Martins

2016–2017 **MSc in Computer Science and Computational Math.**, *University of São Paulo*, USP, SP, Brazil.

Segmentation and Disfluency Detection in Narrative Transcripts from Neuropsychological Tests

**Best MSc thesis at PROPOR 2018.*

Supervisor Dra. Sandra M. Aluísio

Selected Publications (* denotes equal contribution)

- DASHATTENTION: DIFFERENTIABLE AND ADAPTIVE SPARSE HIERARCHICAL ATTENTION
Yuxiang Huang*, Nuno Gonçalves*, Federico Alvetreti, Lei Li, Xu Han, Edoardo Ponti, André F. T. Martins, **Marcos Treviso**
arXiv:2605.18753
- ENTMAXKV: SUPPORT-AWARE DECODING FOR ENTMAX ATTENTION
Gonçalo Duarte, Miguel Couceiro, **Marcos Treviso**
arXiv:2605.21649
- ADASPLASH-2: FASTER DIFFERENTIABLE SPARSE ATTENTION
Nuno Gonçalves*, Hugo Pitorro*, Vlad Niculae, Edoardo Ponti, Lei Li, André F. T. Martins, **Marcos Treviso**
At ICML 2026.
- LONG-CONTEXT GENERALIZATION WITH SPARSE ATTENTION
Pavlo Vasylenko, Hugo Pitorro, André F. T. Martins, **Marcos Treviso**
At ICLR 2026.
- ADSPLASH: ADAPTIVE SPARSE FLASH ATTENTION
Nuno Gonçalves, **Marcos Treviso**, André F. T. Martins
*At ICML 2025. *Spotlight (top 1%)*
- LATIM: MEASURING LATENT TOKEN-TO-TOKEN INTERACTIONS IN MAMBA MODELS
Hugo Pitorro, **Marcos Treviso**
*At ACL 2025. *Outstanding Paper Award*
- HOW EFFECTIVE ARE STATE SPACE MODELS FOR MACHINE TRANSLATION?
Hugo Pitorro*, Pavlo Vasylenko*, **Marcos Treviso**, and André F. T. Martins

At WMT 2024 within EMNLP 2024.

- **xTOWER: A MULTILINGUAL LLM FOR EXPLAINING AND CORRECTING TRANSLATION ERRORS**
Marcos Treviso, Nuno Guerreiro, Sweta Agrawal, Ricardo Rei, José Pombal, Tania Vaz, Helena Wu, Beatriz Silva, Daan van Stigt, and André F. T. Martins
At EMNLP Findings 2024.
- **EFFICIENT METHODS FOR NATURAL LANGUAGE PROCESSING: A SURVEY**
Marcos Treviso*, Ji-Ung Lee*, Tianchu Ji*, Betty Aken, Qingqing Cao, Manuel R Ciosici, Michael Hassid, Kenneth Heafield, Sara Hooker, Pedro H Martins, André Martins, Peter Milder, Colin Raffel, Edwin Simpson, Noam Slonim, Niranjan Balasubramanian, Leon Derczynski, Roy Schwartz
At TACL 2023. Presented at ACL 2023.
- **A JOINT FRAMEWORK FOR RATIONALIZATION AND COUNTERFACTUAL TEXT GENERATION**
Marcos Treviso, Alexis Ross, Nuno Guerreiro, and André F. T. Martins
At ACL 2023.
- **COMETKIWI: IST-UNBABEL 2022 SUBMISSION FOR THE QE SHARED TASK**
Ricardo Rei*, **Marcos Treviso***, Nuno Guerreiro*, Chrysoula Zerva*, Ana C Farinha, Christine Maroti, José GC Souza, Taisiya Glushkova, Duarte M Alves, Alon Lavie, and André F. T. Martins
At WMT within EMNLP 2022. **Shared Task Winner*
- **LEARNING TO SCAFFOLD: OPTIMIZING MODEL EXPLANATIONS FOR TEACHING**
Patrick Fernandes*, **Marcos Treviso***, Danish Pruthi, Graham Neubig, and André F. T. Martins
At NeurIPS 2022.
- **PREDICTING ATTENTION SPARSITY IN TRANSFORMERS**
Marcos Treviso, António Góis, Patrick Fernandes, Erick Fonseca, and André F. T. Martins
At SPNLP within ACL 2022.
- **TOWARDS PREDICTION EXPLAINABILITY THROUGH SPARSE COMMUNICATION**
Marcos Treviso, and André F. T. Martins
At BlackboxNLP within EMNLP 2020.
- **SPARSE AND CONTINUOUS ATTENTION MECHANISMS**
André F. T. Martins, António Farinhas, **Marcos Treviso**, Vlad Niculae, Pedro Aguiar, and Mario Figueiredo
At NeurIPS 2020. **Spotlight Presentation*
- **OPENKIWI: AN OPEN SOURCE FRAMEWORK FOR QUALITY ESTIMATION**
Fabio Kepler, Jonay Trénous, **Marcos Treviso**, Miguel Vera, and André F. T. Martins
At ACL 2019. **Best Demo Paper Award*

Experience

- 2019-2025 **Reviewer** at ACL, EACL, NAACL, EMNLP*, NeurIPS*, ICLR*, ICML*. (*outstanding)
- 2024-2025 **Area chair** at ACL, EACL, and EMNLP.
- 2025- **Senior Area chair** at ACL.
- 06–09/2018 **Junior Research AI Intern** at UNBABEL, Lisbon, Portugal.
I contributed to OpenKiwi, an open source tool for machine translation quality estimation.

Skills

- Languages B. Portuguese (native), English (fluent)
- Technical PYTHON, C, C++, PHP, HTML, CSS, JAVASCRIPT, L^AT_EX, LINUX
↔ PyTorch, JAX, Triton, HuggingFace's Transformers & Datasets & Accelerate, vLLM, PyG, Scikit-Learn, Pandas, Numpy, Scipy, Matplotlib, Seaborn
- Misc. Tensor Puzzles with Sasha Rush.